

Use the formulas, $A = P(1 + r)^t$, $A = P(1 + \frac{r}{n})^{nt}$, or $A = Pe^{rt}$ to problems 1–6.

1. If the inflation rate is 6%, how much will a house currently worth \$465,000 be worth in 10 years?

2. John invests \$1,000 in a mutual fund which grows a rate of 5% compounded semiannually. Find how much the fund is worth after 10 years.

3. John invests \$1,000 in a mutual fund which grows a rate of 5% compounded monthly. Find how much the fund is worth after 10 years.

4. John invests \$1,000 in a mutual fund which grows a rate of 5% compounded daily. Find how much the fund is worth after 10 years.

5. John invests \$1,000 in a mutual fund which grows a rate of 5% compounded continuously. Find how much the fund is worth after 10 years.

6. A city is growing at the rate of 0.6% annually. If there were 3,000,000 residents in 1994, find how many (to the nearest ten-thousand) are living in that city in 2000.